# TriAlign

Progress in comparing clinical trial eligibility criteria

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# TriAlign has two tasks

Help a clinician compare the eligibility requirements of Trial A and Trial B.

**1. Pair identification**

Which criteria should be compared?

- Find candidate counterparts across trials.
- Baselines: TF–IDF, MPNet and BioLORD.
- Output: ranked candidate pairs.

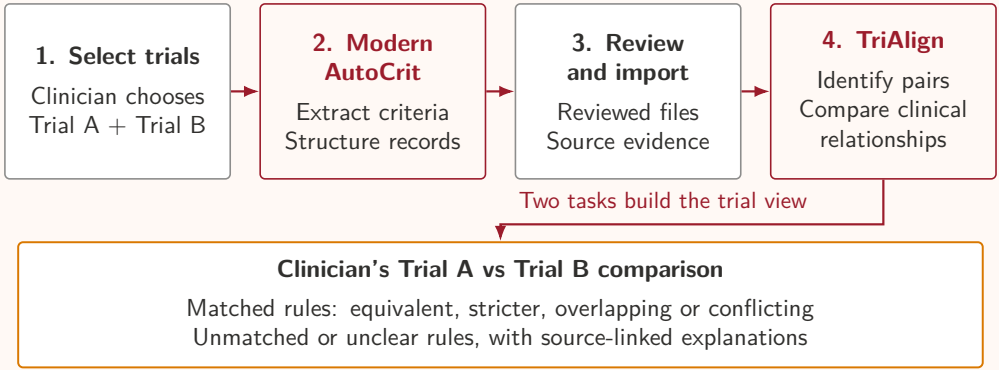
**2. Relationship comparison**

How do those criteria relate clinically?

- Judge equivalence, restrictiveness, overlap and other relationships.
- Exact tools and LLM judge/critic now; trained relationship model planned.
- Output: relationship, evidence or review.

**Example:** identify "age $\geq 18$" and "age $\geq 21$" as counterparts; then determine that $\geq 21$ is more restrictive.

# From extracted criteria to trial comparison




**Demo assumption:** criteria already come from Modern AutoCrit.

**Planned:** extraction directly within TriAlign, while retaining file import.

# Current dataset: already extracted and reviewed

| Data source | Public oncology and cardiovascular trial eligibility text |
|---|---|
| Extraction | Modern AutoCrit produces atomic criteria with structured fields |
| Manual review | Reviewed by the project owner in `trial100_v1.xlsx` |
| Frozen corpus | 3,537 criteria across 100 trials, stored as Parquet |
| Evidence retained | Criterion text, parent statements, trial IDs and source context |

**What the demo loads:** select two trials from this frozen corpus, or import other extracted criterion records.

The demo compares existing extracted criteria. Direct extraction within TriAlign is planned.

# Dataset preview: reviewed criterion records

| | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | source | parent_statement | criterion_type | domain | entity | comparator | value | unit | value_kind | lower_bound | lower_inclusive | upper_bound | upper_inclusive | negated | temporal | repetition | qualifier | logical_operator | review_status |
| 2 | Aged 18 - 80 years old | Aged 18 - 80 years old | inclusion | person | Age | | 18 to 80 | years | interval | 18 | ☑ | 80 | ☑ | ☐ | | | | standalone | corrected |
| 3 | Able to give informed c | Able to give informed consent | inclusion | person | Informed Consent | | present | | categorical | | | | | ☐ | | | | standalone | corrected |
| 4 | Available for the whole | Available for the whole durat | inclusion | observation | availability for study duration | | present | | categorical | | | | | ☐ | | | | standalone | corrected |
| 5 | Life expectancy \>2 yea | Life expectancy \>2 years, in t | inclusion | observation | Life Expectancy | > | 2 | years | numerical | 2 | ☐ | | | ☐ | | | | standalone | corrected |
| 6 | Current systemic thera | Current systemic therapy for t | exclusion | drug | current systemic therapy for bladc | | present | | categorical | | | | | ☐ | | | | standalone | corrected |
| 7 | History of malignancy o | History of malignancy of othe | exclusion | condition | malignancy of other organ system | | present | | categorical | | | | | ☐ | History of; within past 5 years; at least 24 months | | | or | corrected |
| 8 | History of malignancy o | History of malignancy of othe | inclusion | measurement | time since nephroureterecto | >= | 24 | months | numerical | 24 | ☑ | | | ☐ | after nephroureterectomy | | exception to other-organ r | or | corrected |
| 9 | Eastern Cooperative On | Eastern Cooperative Oncolog | inclusion | observation | ECOG Performance Status | | 0 to 3 | | interval | 0 | ☑ | 3 | ☑ | ☐ | | | | standalone | corrected |
| 10 | Serum creatinine ≤ 2.0 ı | Serum creatinine ≤ 2.0 mg/dl. | inclusion | measurement | Serum Creatinine | <= | 2 | mg/dl | numerical | | | 2 | ☑ | ☐ | | | | standalone | corrected |
| 11 | Subjects with detectabl | Subjects with detectable level | exclusion | condition | detectable levels of endogenous ar | | present | | categorical | | | | | ☐ | | | at the time of screening | standalone | corrected |
| 12 | In the absence of rapidl | In the absence of rapidly prog | inclusion | measurement | time since prior cytotoxic tr | >= | 2 | weeks | numerical | 2 | ☑ | | | ☐ | before study drug administration | | in the absence of rapidly p | or | corrected |
| 13 | Histologically confirme | Histologically confirmed new | inclusion | condition | GBM | | present | | categorical | | | | | ☐ | | | | standalone | corrected |
| 14 | Supratentorial tumor lc | Supratentorial tumor locatio | inclusion | condition | supratentorial tumor location | | present | | categorical | | | | | ☐ | | | | standalone | corrected |
| 15 | Planned treatment with | Planned treatment with adju | inclusion | drug | TMZ | | 150 to 200 | mg/m^2 dail | interval | 150 | ☑ | 200 | ☑ | ☐ | | | | standalone | corrected |
| 16 | All patients with sufficie | All patients with sufficient tis | inclusion | observation | mgmt promoter methylation dete | | present | | categorical | | | | | ☐ | prior to enrollment | | | standalone | corrected |
| 17 | Karnofsky Performance | Karnofsky Performance Status | inclusion | measurement | Karnofsky Performance Stat | >= | 70 | | numerical | 70 | ☑ | | | ☐ | | | | standalone | corrected |
| 18 | Are surgically sterile, OR | Are surgically sterile, OR | inclusion | person | surgically sterile | | present | | categorical | | | | | ☐ | | | | or | corrected |
| 19 | Unstable angina at regis | Unstable angina at registratio | exclusion | condition | unstable angina | | present | | categorical | | | | | ☐ | at registration | | | standalone | corrected |
| 20 | Transmural myocardial | Transmural myocardial infarc | exclusion | condition | transmural myocardial infarction | | present | | categorical | | | | | ☐ | within the last six months prior to registration | | | standalone | corrected |
| 21 | Patents treated on any ı | Patents treated on any other t | exclusion | person | patients treated on any other ther | | present | | categorical | | | | | ☐ | within 30 days prior to registration | | | standalone | corrected |
| 22 | Inability to undergo MF | Inability to undergo MRI. | exclusion | device | inability to undergo mri | | present | | categorical | | | | | ☐ | | | | standalone | corrected |
| 23 | Are postmenopausal for | Are postmenopausal for at lea | inclusion | measurement | duration of postmenopausal | >= | 1 | year | numerical | 1 | ☑ | | | ☐ | before the screening visit | | deterministically parsed v | or | corrected |
| 24 | Previously treated with | Previously treated with brent | exclusion | drug | brentuximab vedotin | | present | | categorical | | | | | ☐ | | | | or | corrected |
| 25 | Previously treated with | Previously treated with brent | exclusion | drug | immune-oncology agents | | present | | categorical | | | | | ☐ | | | | or | corrected |
| 26 | Previously treated with | Previously treated with brent | exclusion | procedure | allogeneic stem cell transplant | | present | | categorical | | | | | ☐ | | | | or | corrected |
| 27 | Previously treated with | Previously treated with brent | exclusion | procedure | autologous stem cell transplant | | present | | categorical | | | | | ☐ | | | | or | corrected |
| 28 | Documented history of a | Documented history of a care | exclusion | observation | cerebral vascular event | | present | | categorical | | | | | ☐ | history of | | | standalone | corrected |
| 29 | History of progressive n | History of progressive multifo | exclusion | condition | progressive multifocal leukoence | | present | | categorical | | | | | ☐ | History of | | | standalone | corrected |
| 30 | Hemoglobin ≥ 8.0 g/dl | Hemoglobin ≥ 8.0 g/dl | inclusion | measurement | Hemoglobin | >= | 8 | g/dl | numerical | 8 | ☑ | | | ☐ | | | | standalone | corrected |
| 31 | Absolute neutrophil co | Absolute neutrophil count ≥ 1 | inclusion | measurement | Absolute Neutrophil Count | >= | 1000 | mcL | numerical | 1000 | ☑ | | | ☐ | | | | standalone | corrected |
| 32 | Platelet count ≥ 75,000 | Platelet count ≥ 75,000/mcL | inclusion | measurement | Platelet Count | >= | 75000 | mcL | numerical | 75000 | ☑ | | | ☐ | | | | standalone | corrected |
| 33 | Aspartate transaminase | Aspartate transaminase (AST) | inclusion | measurement | Aspartate Aminotransferase | <= | 2.5 | | numerical | | | 2.5 | ☑ | ☐ | | | | standalone | corrected |
| 34 | Alanine transaminase (A | Alanine transaminase (ALT) (SC | inclusion | measurement | Alanine Aminotransferase (A | <= | 2.5 | | numerical | | | 2.5 | ☑ | ☐ | | | | standalone | corrected |
| 35 | International normalize | International normalized ratio | inclusion | measurement | INR | <= | 1.5 | | numerical | | | 1.5 | ☑ | ☐ | | | | standalone | corrected |
| 36 | HBsAg negative, HBcAb, | HBsAg negative, HBcAb negat | inclusion | condition | hbsab positive | | present | | categorical | | | | | ☐ | | | applies to patients with HI | standalone | corrected |

Each row retains the criterion text and parent statement alongside its role, clinical concept, numeric bounds, timing, qualifiers and review status.

# Task 1: pair identification baselines

**Jiatong: baseline pair ranking**

- TF–IDF, MPNet and BioLORD score and rank candidate pairs.
- Cross-trial ranking with a shared neighbor format.
- Provisional runs on 200 criteria across 100 trials.

**Current local demo**

- A clinical-concept gate and TF–IDF ranking select one-to-one candidate pairs.
- Selected candidate pairs pass to Task 2 for clinical relationship comparison.
- Dense baseline integration awaits the teammate PR.

**Training plan:** improve pair identification with a clinical encoder, and learn directional relationships with a separate prediction head.

Evaluate the two tasks separately: finding the right counterparts and assigning the right relationships.

# Task 2: observed relationship comparison

**16/16 pairs examined: 9 supported proposals; 7 need review**

**NCT01841736 · 104 criteria**

"Women must not be pregnant or nursing"

**NCT04310007 · 136 criteria**

"Patients must not be pregnant or breast-feeding"

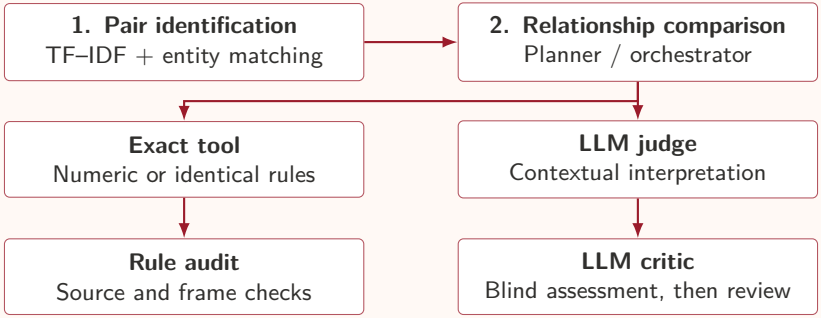
## Pregnancy restrictions: proposed equivalent

Both trials exclude pregnant participants. Equivalence assumes comparable assessment at enrollment. Breastfeeding is compared separately.

**Unresolved example: platelets.** The critic rejected a directional label because Trial A's measurement timing was unspecified. Omitted timing does not establish a less restrictive rule.

One demonstration run. Agent acceptance is not clinical validation; accepted labels still require review.

# The two tasks in the agentic workflow




**Outcome:** an evidence-supported proposal or an unresolved case for human review.

The orchestrator allows one revision, then finalizes or escalates. Retrieval similarity alone never supplies a clinical label.

# Skills, tools and hooks give the agents boundaries

| Component | Role in the working system |
|---|---|
| Skills | Versioned instructions for numeric comparison, clinical context and source problems. The planner selects a skill. |
| Tools | Inspect the comparison frame, compare supported constraints and collect source citations. Only permitted tools run. |
| Hooks | Validate inputs and exact source quotes, check limits and cancellation, and gate the final result. |
| Orchestrator | Coordinates tool calls, judge, critic and revision within pair, call and time limits. |

**Clinician-facing output:** source excerpts, rationale, critic feedback, executed trace and saved reviewer decisions.

The critic first assesses sources blind, then reviews the judge. Models are separately configurable; agreement is not clinical validation.

# Closest related methods and benchmarks

| Related work | Overlap with TriAlign | Planned comparison |
|---|---|---|
| CReSE | Criterion embeddings and clinical relevance | Task 1: rank counterpart criteria. |
| EligMeta | Eligibility alignment to a target trial | Task 2: compare supported criterion differences. |
| NLI4CT (benchmark) | Clinical-trial entailment and contradiction | Task 2: stress-test reasoning and consistency. |
| AMEND++ (benchmark) | Eligibility amendment prediction and version histories | Adapt substantive edits for semantic-change tests. |

Planned comparisons, subject to artifact access. Native outputs differ from TriAlign's relationship labels, so matched cases or explicit adaptations are required.

# Current challenges and limitations

- Observed failures include source-quotation mismatches and unsupported timing assumptions, including in some accepted overlap proposals.
- Judge and critic can share mistakes. Human review remains necessary.
- No retrieved candidate does not prove that a requirement is absent.
- Independent clinical evaluation, final family-aware splits and calibrated uncertainty remain pending.
- Dense baseline integration and fine-tuning remain ahead.

The UI runs locally. Contextual model calls still need network access and have API costs and latency.

# Next steps before the final presentation

| Task | Deliverable |
|---|---|
| Pair construction | Expand candidate pairs using structured contrasts and baseline-ranked counterparts. |
| Clinical review | Adjudicate relationships and governing context; obtain independent review of a subset. |
| Evaluation partitions | Freeze family-aware train, development and test splits. |
| Baseline integration | Integrate MPNet/BioLORD; reproduce full-corpus rankings with pinned revisions. |
| Training and evaluation | Train the encoder and relationship head when labels are ready. Evaluate both tasks separately, including errors and abstention. |
| Local demonstration | Refine import/review flows, measure latency and rehearse locally. |

# Pair construction and family-aware sampling

**Built: 75 cross-trial pilot pairs**

| Sampling stratum | Pairs |
|---|---|
| Same entity, structured differences | 25 |
| Other same-entity pairs | 25 |
| Lexical neighbors | 15 |
| Random controls | 10 |

**Family-aware split preparation**

- Audited trial identity, lineage and copied criteria across 100 trials.
- Reserved pilot trials for development; 21 trials excluded from held-out testing.
- Keep linked trials together to limit leakage. Final train/dev/test assignments are pending.

**Next sampling:** freeze family-aware partitions first, then sample same-concept and TF–IDF/MPNet/BioLORD-ranked candidates within each partition. Include difficult scope/negation cases and random controls; cap reuse.

Sampling proposes pairs. Clinical review supplies the relationship labels.